

CROSS-E-L3-G-CKN-01 v1: Cross Eweak + L3 + G-CKN concentration

ZFL Lab

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INPUT: $C_E^{weak} = \text{PROVEN}$: $T_* < \infty \implies \limsup_{t \rightarrow T_*} \int_0^t W = +\infty$, $W = \int \omega^T S \omega$. $C_{L^3} = \text{PROVEN}$: $T_* < \infty \implies \limsup_{t \rightarrow T_*} \|u(t)\|_3 = +\infty$. $C_G^{CKN} = \text{PROVEN-CONDITIONAL}$ ON CKN-1982: $\exists(x_*, T_*)$ singular, $\forall r \downarrow 0$ small, $Q(r) := r^{-2} \int_{Q_r(x_*, T_*)} |u|^3 + |p|^{3/2} \geq \epsilon_{CKN}$, $Q_r = B_r \times (T_* - r^2, T_*)$, ϵ_{CKN} universal. $C_G^{L^3} = \text{OPEN}$: $\exists t_n, x_n, r_n \downarrow 0 : \|u(t_n)\|_{L^3(B(x_n, r_n))} \geq \epsilon_0$.

FORBIDDEN: $C_G^{CKN} \Rightarrow C_G^{L^3}$ without bridge. $C_{L^3} \Rightarrow C_G^{CKN}$ without bridge. CKN $\Rightarrow$ contradiction. Conditional result counted as PROVEN.

MISSION: Cross $C_E^{weak} + C_{L^3} + C_G^{CKN}$.

AUDIT:

1. Logical compatibility:

C_E^{weak} : space integral of stretching sign indefinite, cumulative diverging in limsup. C_{L^3} : global critical norm diverging in limsup. C_G^{CKN} : local space-time + pressure quantity bounded below near singular point.

No direct logical incompatibility: can all hold simultaneously for same singular solution.

$$\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_G^{CKN}}$$

2. Implication bridges:

a) $C_E^{weak} \wedge C_{L^3} \Rightarrow C_G^{CKN}$? C_E^{weak} from enstrophy, C_{L^3} from ESS. C_G^{CKN} requires CKN epsilon-regularity contrapositive. No theorem in $\mathcal{C}_{PROVEN}$ gives CKN. So implication NOT PROVEN without EXTERNAL CKN.

$$\boxed{C_E^{weak} \wedge C_{L^3} \not\Rightarrow C_G^{CKN} \text{ without CKN dependency}}$$

b) $C_G^{CKN} \wedge C_{L^3} \Rightarrow C_E^{weak}$? C_G^{CKN} gives local concentration, C_{L^3} global. C_E^{weak} is necessary from E-04. But $C_G^{CKN} \wedge C_{L^3}$ alone does not produce W divergence. No bridge.

$$\boxed{C_G^{CKN} \wedge C_{L^3} \not\Rightarrow C_E^{weak} \text{ without E-04}}$$

c) $C_G^{CKN} \Rightarrow C_E^{weak}$? No. C_G^{CKN} controls u, p , not W sign.

Thus no circular implication proven.

3. Scaling compatibility:

W : scaling $[time]^{-3}$, $\int W \sim$ enstrophy growth. $\|u\|_3$: critical invariant. $Q(r)$: critical invariant $r^{-2} \int_{Q_r} |u|^3$. All scale-critical, so compatible in scaling.

$$\boxed{\text{Scaling COMPATIBLE}}$$

4. CKN dependency propagation:

C_G^{CKN} remains CONDITIONAL. Cannot be elevated to PROVEN in $\mathcal{C}_{PROVEN}$. If we use C_G^{CKN} in further CROSS, output must retain tag EXTERNAL-DEPENDENCY: CKN-1982.

$$\boxed{C_G^{CKN} = \text{CONDITIONAL}}$$

5. $\perp$?

No contradiction derived from conjunction.

$$\boxed{\text{No } \perp \text{ from } C_E^{weak} \wedge C_{L^3} \wedge C_G^{CKN}}$$

OUTPUT:

$$\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_G^{CKN} = \text{COMPATIBLE} / \text{OPEN}}$$

$$\boxed{C_G^{L^3} \neq C_G^{CKN} \text{ maintained}}$$

C_G^{CKN} CONDITIONAL ON CKN, not promoted

Interpretation: Three singularity signals are mutually compatible. Bottleneck for contradiction lies in fourth structure: vorticity direction, pressure depletion, geometric alignment, or stronger W_- control (E-strong-neg-02 blocked). No new PROVEN condition added to $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$.

Next front: geometry / depletion, not more conditions.